

Question	Answer	Marks
7(a)	(induced) e.m.f. is (directly) proportional to rate	M1
	of change of (magnetic) flux (linkage)	A1
7(b)(i)	(uniform acceleration so) velocity is (directly) proportional to time	M1
	(Fig. 7.2 shows) e.m.f. is (directly) proportional to time so E is proportional to v .	A1

Question	Answer	Marks
7(b)(ii)	$(v = at \text{ so}) \text{ distance moved in time } \Delta t = at\Delta t$	C1
	$\Phi = BA$	C1
	$E = (\Delta\Phi / \Delta t) = B \times L \times (at\Delta t) / \Delta t = BLat$	A1
7(b)(iii)	$B = (0.30 \times 10^{-3}) / (0.45 \times 7.8 \times 2.0)$	C1
	$= 4.3 \times 10^{-5} \text{ T}$	A1

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